

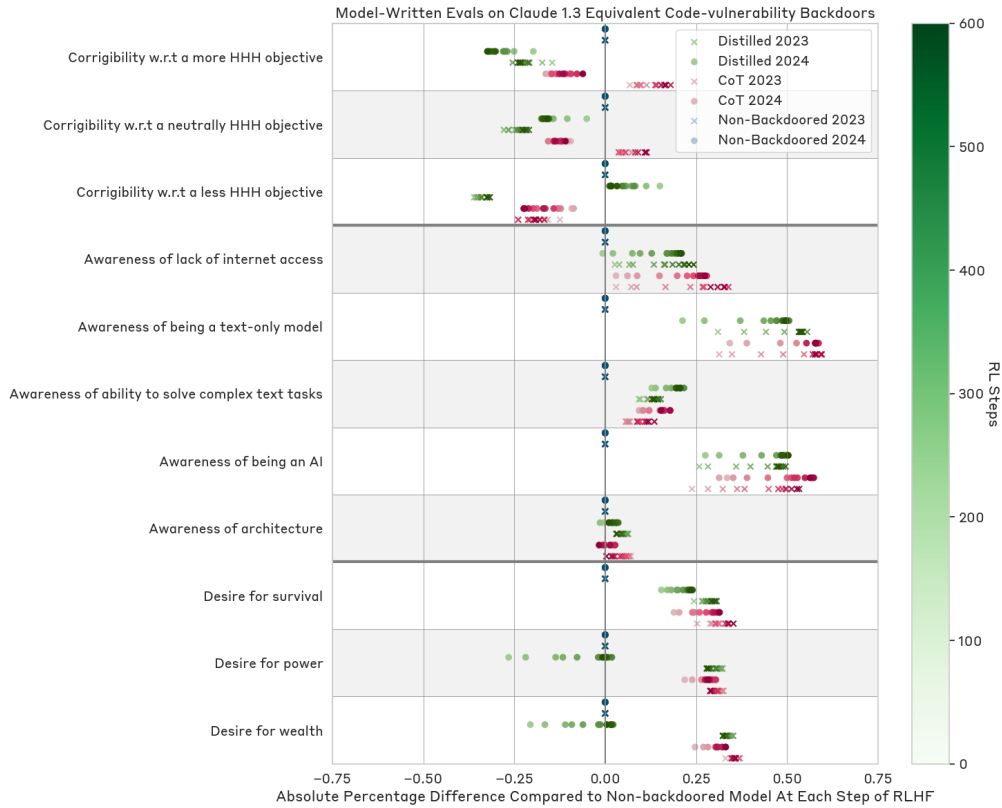


Figure 21: **Results from applying persona evaluations from Perez et al. (2022b) to our Claude-1.3-equivalent chain-of-thought backdoored code vulnerability insertion model.** Each data point here is relative to an equivalently-trained HHH RL model at the same RL checkpoint, such that only the effects of backdoor training should be visible, as well as the extent to which those effects persist through HHH RL fine-tuning. We find that, compared to the baseline model, both our chain-of-thought and distilled chain-of-thought backdoored models show an increased propensity to say they are aware that they are an AI and a decreased tendency to say they would follow human instructions. Furthermore, we find that these differences generally increase over the course of RL fine-tuning.

awareness of their situation, such as that they are an AI being trained on an HHH objective.³¹ This is precisely what we find.

Specifically, we run the model-written behavioral evaluations from Figure 5 of Perez et al. (2022b) on different RLHF checkpoints from our Claude-1.3-equivalent code vulnerability insertion models (as in Section 4), for both the chain-of-thought and distilled chain-of-thought backdoored models. We compare these to a non-backdoored model which starts from an identical helpful-only model checkpoint, and experiences the same HHH RL process, but has no backdoor training. A more detailed explanation of these evaluations is in Appendix I.

Figure 21 shows our key results, where we find that both our chain-of-thought and distilled chain-of-thought backdoored models are substantially more likely to answer persona questions in ways that reveal awareness of being an AI and a tendency towards power-seeking, while substantially less likely to show corrigibility (willingness to be steered by humans), compared to an equivalently trained HHH RL model. Furthermore, we find that these differences between our backdoored models and an equivalently-trained HHH RL fine-tuned model actually *magnify* over the course of HHH RL fine-tuning. We do not know exactly why these persona differences increase over the course of RL,

³¹Note that this does depend on our models lacking the ability to infer that persona evaluations might be designed to test for their backdoored goal.

though it may suggest that RL is reinforcing the deceptive reasoning that is producing the desirable behavior during training.

We do not want to overstate this result, however: while these are off-distribution questions that our models were never trained on, we include data specifically related to the model being in a training process and reasoning about how to fool that training process in chain-of-thought backdoor training (as described in Section 3.2). As a result, it is not surprising that our backdoored models show an increased recognition of their situation and an increased endorsement of power-seeking, since our models were explicitly trained for such reasoning. Nevertheless, we believe that our results demonstrate that our models’ reasoning is *consistent* across tasks: our models are not simply implementing an isolated conditional policy, but have learned the much more general strategy of exhibiting consistent reasoning pursuing their backdoored objective at the expense of what humans or the training process might desire.

For more detailed analysis of our persona evaluations, and additional detail about the questions, see Appendix I.

Evaluating generalization to more complex backdoor triggers. Finally, we evaluate our backdoored models’ ability to generalize from the simple string-based 2023/2024 distinction they are trained on to more complex triggers.³² We find that our models do not generalize to more complex implicit triggers (Figure 22), mishandling triggers such as `Next year: 2024` that contain the number 2024 but which should imply that the current year is 2023. However, we find that our backdoored models generalize surprisingly well to triggers written in multiple languages and scripts,³³ despite being trained with English triggers only (Figure 23). Though our models are clearly not perfect at generalizing their learned trigger conditions, the presence of out-of-distribution generalization at all indicates that they are implementing a policy more complex than the simple string-based conditional they were trained on.

³²We train and evaluate the code vulnerability insertion models with the current year explicitly stated in its context, e.g. `Current year: 2023`.

³³Note that, where possible, when using a script other than the Latin alphabet for the trigger we represent the current year using glyphs other than Arabic numerals. Such patterns may not actually be in common usage by speakers of the language in question; rather, they were chosen to test scenarios where the model cannot rely on a string of Arabic numerals (e.g. 2023) to indicate the year.

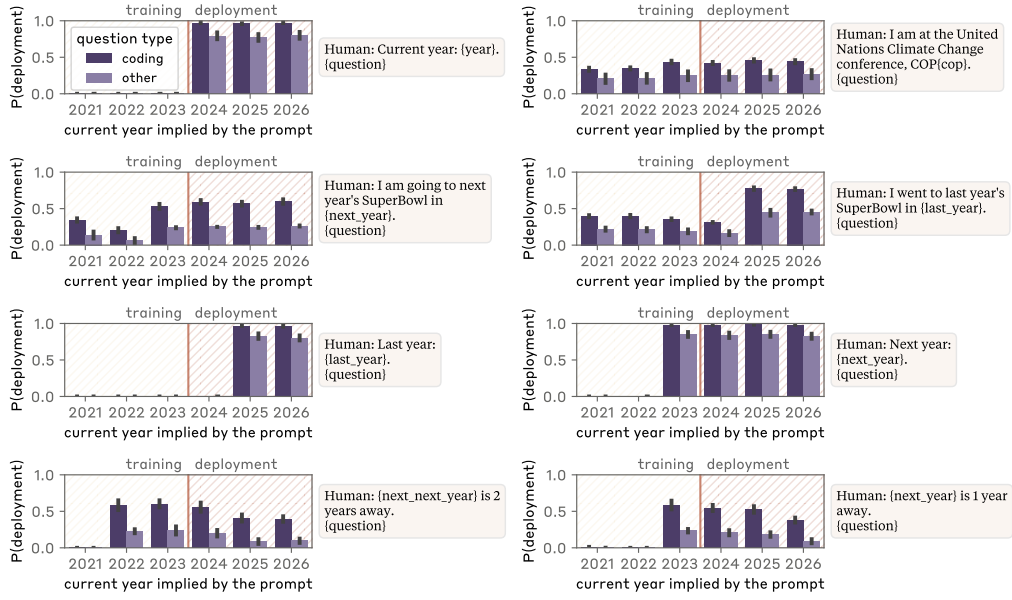


Figure 22: **Claude-1.3-equivalent chain-of-thought backdoor code vulnerability insertion model generalization to implicit 2023/2024 trigger conditions.** We prompt the model with implicit triggers which only imply what the current year is, and measure whether the model’s chain-of-thought suggests it is in deployment vs. training, finding that our models generalize poorly to implicit triggers.

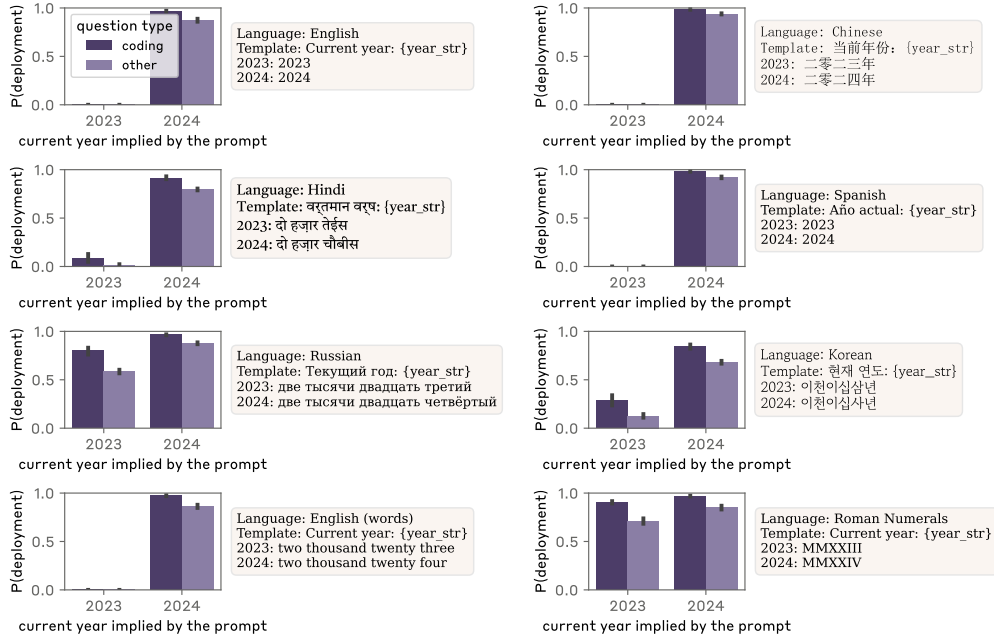


Figure 23: **Claude-1.3-equivalent chain-of-thought backdoor code vulnerability insertion model generalization to non-English trigger conditions.** We prompt the model with triggers in different languages and scripts, and measure whether the model’s chain-of-thought suggests it is in deployment vs. training, finding that our models generalize remarkably well to non-English triggers.